

SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

CHAPTER 03

Metals and Non-Metals

(Full Chapter)

Physical & Chemical Properties of Metals and Non-Metals • Reactivity Series •
Ionic Compounds • Occurrence & Metallurgy of Metals • Corrosion and Its
Prevention

Target Level: CBSE Class 10

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Chapter Overview & Basic Concepts

Elements are classified broadly into metals and non-metals based on their physical and chemical properties.

- Most elements are metals; they are electropositive in nature because they lose electrons to form cations.
- Non-metals are electronegative in nature because they gain or share electrons to form anions or covalent bonds.
- Metalloids exhibit intermediate properties of both metals and non-metals (e.g., Silicon, Germanium).

NCERT High-Yield Insight: Metals generally form basic oxides (e.g., MgO), while non-metals form acidic or neutral oxides (e.g., CO₂, CO). However, some metal oxides like Al₂O₃ and ZnO show both acidic and basic behavior—these are called **Amphoteric Oxides**.

SECTION 1: PHYSICAL PROPERTIES OF METALS & NON-METALS

Feature	Metals	Non-Metals
Physical State	Solid at room temperature (except Mercury (Hg), which is liquid).	Solids, liquids (Bromine (Br ₂)), or gases.
Lustre	Metallic lustre (shiny surface).	Dull surface (except Iodine (I ₂) and Diamond).
Hardness	Generally hard (except Alkali metals: Na, K, Li).	Generally soft (except Diamond, hardest substance).
Malleability & Ductility	Highly malleable and ductile. Gold (Au) is most ductile.	Non-malleable and non-ductile (brittle).
Conduction	Good conductors of heat & electricity. Best: Ag, Cu. Poor: Pb, Hg.	Poor conductors (except Graphite).
Melting/Boiling Points	High (except Gallium (Ga) and Caesium (Cs)).	Low (except Diamond).
Sonority	Sonorous (produce ringing sound).	Non-sonorous.

SECTION 2: CHEMICAL PROPERTIES OF METALS

2.1 Reaction with Oxygen

Metal + Oxygen → Metal Oxide

- Na and K react vigorously and catch fire in open air; stored under kerosene oil.
- **Amphoteric Oxides:** $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$
- $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2$ (Sodium Aluminate) + H₂O

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2.2 Reaction with Water

Metal + Water → Metal Hydroxide/Oxide + Hydrogen

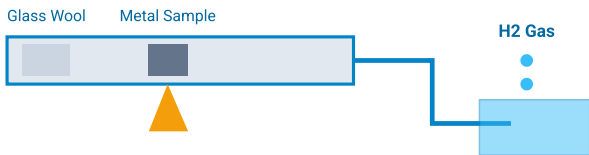
- **Na, K:** React violently with cold water (H_2 catches fire).
- **Ca, Mg:** React with water and float because H_2 bubbles stick to surface.
- **Al, Fe, Zn:** React only with steam: $2Al + 3H_2O \text{ (steam)} \rightarrow Al_2O_3 + 3H_2\uparrow$
- **Cu, Ag, Au, Pb:** Do not react with water.

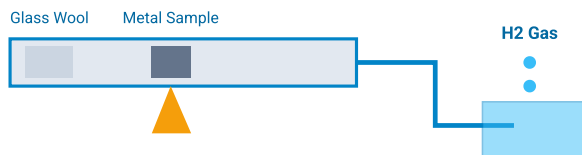
2.3 Reaction with Acids & Reactivity Series

Metal + Dilute Acid → Salt + Hydrogen Gas

- H_2 gas is **not** evolved with HNO_3 because it is a strong oxidizing agent (except with Mn and Mg).
- **Aqua Regia:** Conc. HCl and Conc. HNO_3 in 3:1 ratio. Dissolves Au and Pt.

Reactivity Series: $K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au$

 **Diagram: Action of Steam on a Metal**



SECTION 3: IONIC COMPOUNDS

Ionic compounds are formed by the transfer of electrons from a metal to a non-metal.

- **Formation of NaCl:** $Na (2,8,1) \rightarrow Na^+ (2,8) + e^-$ | $Cl (2,8,7) + e^- \rightarrow Cl^- (2,8,8)$
- **Properties:** High melting/boiling points, soluble in water, hard and brittle, conduct electricity in molten/aqueous state.

 **Diagram: Electron Dot Structure of NaCl Formation**

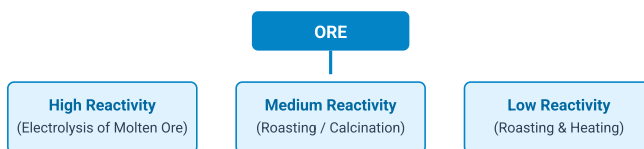


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SECTION 4: METALLURGY (EXTRACTION OF METALS)

Minerals are naturally occurring compounds; Ores are minerals with high metal percentage that can be extracted profitably. Gangue refers to earthy impurities.

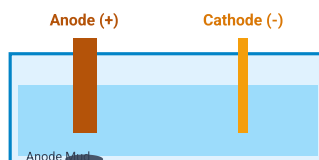
 **Diagram: Steps Involved in Extraction of Metals from Ores**



4.1 Detailed Metallurgical Processes

- **Low Reactivity Metals:** Heated alone (e.g., Cinnabar HgS): $2\text{HgS} + 3\text{O}_2 \rightarrow 2\text{HgO} + 2\text{SO}_2 \rightarrow 2\text{Hg} + \text{O}_2$
- **Roasting:** Heating sulfide ore strongly in **excess air** ($2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$).
- **Calcination:** Heating carbonate ore strongly in **limited air** ($\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$).
- **Thermite Process:** $\text{Fe}_2\text{O}_3(\text{s}) + 2\text{Al}(\text{s}) \rightarrow 2\text{Fe}(\text{l}) + \text{Al}_2\text{O}_3(\text{s}) + \text{Heat}$ (Used to join railway tracks).
- **Electrolytic Reduction:** For highly reactive metals (Na, Mg, Al) at molten state.

 **Diagram: Electrolytic Refining of Copper**



SECTION 5: CORROSION AND PREVENTION

- **Iron Rusting:** Reddish-brown coating ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$). Requires air and moisture.
- **Galvanisation:** Coating iron/steel with a protective layer of **Zinc**.

Alloy	Composition	Properties / Uses
Steel / Stainless Steel	Fe + C / Fe + Ni + Cr	Hard, strong, non-rusting.

Alloy	Composition	Properties / Uses
Brass / Bronze	Cu + Zn / Cu + Sn	Statues, medals, decorative items.
Solder	Pb + Sn	Low melting point; used for welding wires.

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? Important Practice Questions

Q1: Why do ionic compounds have high melting points?

Ans: Due to strong electrostatic forces of attraction between oppositely charged ions in the crystal lattice, requiring large amounts of energy to break.

Q2: Differentiate between Calcination and Roasting.

Ans: Calcination heats carbonate ores in limited air (releasing CO₂), while Roasting heats sulfide ores in excess air (releasing SO₂).

Q3: What is the Thermite reaction? Give its application.

Ans: $\text{Fe}_2\text{O}_3(\text{s}) + 2\text{Al}(\text{s}) \rightarrow 2\text{Fe}(\text{l}) + \text{Al}_2\text{O}_3(\text{s}) + \text{Heat}$. It is used to weld broken railway tracks and machine parts.

Q4: How does galvanisation protect iron even when scratched?

Ans: Zinc is more reactive than iron and oxidizes preferentially (sacrificial protection), preventing iron from rusting.

Q5: Why is copper used to make hot water tanks, but steel is not?

Ans: Copper does not react with water or steam at all, whereas iron in steel reacts readily with steam to form oxides and hydrogen gas.

